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Special Article

CARDIOVASCULAR

Perioperative management of patients with pulmonary hypertension undergoing non-cardiothoracic, non-obstetric surgery: a systematic review and expert consensus statement

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Abstract

Background: The risk of complications, including death, is substantially increased in patients with pulmonary hypertension (PH) undergoing anaesthesia for surgical procedures, especially in those with pulmonary arterial hypertension (PAH) and chronic thromboembolic PH (CTEPH). Sedation also poses a risk to patients with PH. Physiological changes including tachycardia, hypotension, fluid shifts, and an increase in pulmonary vascular resistance (PH crisis) can precipitate acute right ventricular decompensation and death.

Methods: A systematic literature review was performed of studies in patients with PH undergoing non-cardiac and non-obstetric surgery. The management of patients with PH requiring sedation for endoscopy was also reviewed. Using a

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framework of relevant clinical questions, we review the available evidence guiding operative risk, risk assessment, preoperative optimisation, and perioperative management, and identifying areas for future research.
Results: Reported 30 day mortality after non-cardiac and non-obstetric surgery ranges between 2% and 18% in patients with PH undergoing elective procedures, and increases to 15–50% for emergency surgery, with complications and death usually relating to acute right ventricular failure. Risk factors for mortality include procedure-specific and patient-related factors, especially markers of PH severity (e.g. pulmonary haemodynamics, poor exercise performance, and right ventricular dysfunction). Most studies highlight the importance of individualised preoperative risk assessment and optimisation and advanced perioperative planning.
Conclusions: With an increasing number of patients requiring surgery in specialist and non-specialist PH centres, a systematic, evidence-based, multidisciplinary approach is required to minimise complications. Adequate risk stratification and a tailored-individualised perioperative plan is paramount.

Keywords: acute right ventricular failure; endoscopy; perioperative management; pulmonary hypertension; pulmonary hypertensive crisis; sedation

Editor's key points

- Pulmonary hypertension increases the risk of morbidity and mortality in patients undergoing anaesthesia or sedation for surgery and endoscopy.
- A systematic literature review was performed of the impact of pulmonary hypertension on outcomes after general anaesthesia or sedation for non-cardiothoracic or non-obstetric surgery or endoscopy.
- Risk factors for mortality include procedure-specific and patient-related factors, including emergency procedures and markers of pulmonary hypertension severity.
- Individualised preoperative risk assessment and treatment optimisation and advanced perioperative planning may improve outcomes.
- A systematic, evidence-based, multidisciplinary approach is required to minimise complications.

Patients with pulmonary hypertension (PH) undergoing general anaesthesia for surgical or other procedures are at increased risk of fatal complications.[1–5] Sympathetic nervous system activation, hypoxaemia, or hypercapnia and the use of mechanical ventilation and PEEP can have disastrous consequences on pulmonary haemodynamics, with rapid onset of acute right ventricular (RV) decompensation and cardiac arrest.[6] Postoperatively, abdominal distension, fluid shifts, and pain can also impact negatively on RV performance and cardiac output. Perioperative mortality remains significantly higher in patients with PH than in patients without PH, even when perioperative care is delivered in expert centres.

Patient selection, consideration of non-operative options, not using general anaesthesia where possible, optimisation of PH medications, appropriate and timely conversion to non-enteral therapies, and intraoperative care by anaesthetists knowledgeable in PH are likely to be essential in improving outcomes in these high risk patients.[4,5,7]

We provide an overview of PH and its pathophysiology in relation to the stress of anaesthesia and surgery. Moreover, we systematically reviewed clinical studies on patients with PH undergoing major surgery, and provide expert recommendations based on current available evidence and expert opinion.

Methods

A systematic literature search for clinical studies published between January 1990 and January 2020 in adult patients with PH undergoing general anaesthesia or sedation was performed using PubMed and EMBASE and scoping of personal reference lists. The terms 'pulmonary hypertension' and 'anaesthesia', 'sedation' or 'endoscopy' were used. Abstracts of non-English papers were screened. A detailed search strategy is provided in Supplementary Table S1. Study flow is shown in Figure 1. From more than 6500 publications identified (PubMed and EMBASE), 1801 abstracts were screened, of which 13 clinical studies of adult patients with PH undergoing surgery and eight studies for patients with PH undergoing endoscopy were identified. In addition, 16 relevant review articles and five case reports were selected for use. Studies of patients with PH undergoing obstetric or cardiothoracic surgery in PH were excluded to focus on the management of general surgery in adult patients with PH. The evidence collected was used to inform recommendations and a specific consensus statement.

Definition and management of pulmonary hypertension

Pulmonary hypertension (PH) is defined at right heart catheterisation (RHC) by a mean pulmonary arterial pressure (mPAP) greater than or equal to 25 mm Hg.[8] In the 2018 PH World Symposium, a reduction in the upper mPAP limit to 20 mm Hg was proposed, as 25 mm Hg represents more than 2 SD above normal (Table 1).[8] Patients with PH are classified into five clinical groups according to haemodynamic, aetiologic, and pathologic characteristics (Table 1).[8]

Pulmonary vasodilator therapy is predominantly indicated for the treatment of patients with Group 1 PH (pulmonary arterial hypertension [PAH]) and Group 4 PH (mainly chronic thromboembolic PH [CTEPH]), although the treatment of choice for CTEPH is pulmonary endarterectomy. Management of PH in patients with Group 2 and 3 PH is, in most cases, focussed on the underlying heart or lung disease.

International PH guidelines strongly recommend that a validated risk assessment score in PAH patients should influence the type and intensity of treatment. The aim is to achieve and maintain a low risk profile for all patients, hence improving survival (Table 2).[9] Transplantation (usually

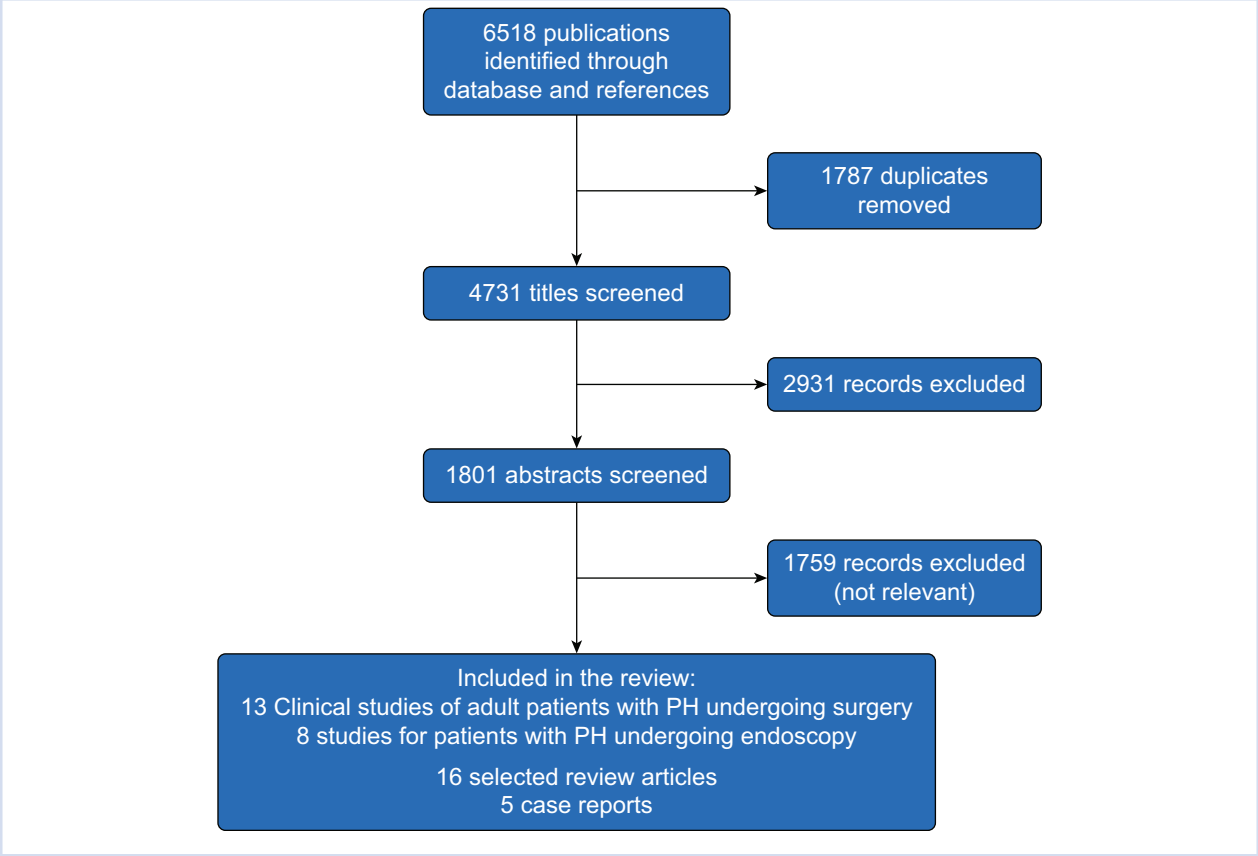


Fig 1. Study flow diagram. Illustration of publications identified using the search terms as defined in the Methods section (see Supplementary Table S1). From more than 6500 publications identified (PubMed and EMBASE), 1801 abstracts were screened, of which 13 clinical studies of adult patients with pulmonary hypertension (PH) undergoing surgery and eight studies for patients with PH undergoing endoscopy were identified. In addition, 16 relevant review articles and five case reports were selected.

Table 1 Updated classification and definitions of pulmonary hypertension. PH, pulmonary hypertension; PAH, pulmonary arterial hypertension; RHC, right heart catheterisation; mPAP, mean pulmonary artery pressure; PCWP, pulmonary capillary wedge pressure; PVR, pulmonary vascular resistance; WU, Wood units; HIV, human immunodeficiency virus.

Classification group	
Group 1	Pulmonary arterial hypertension (PAH). Idiopathic PAH, connective tissue disease associated PAH, and congenital heart disease (mainly Eisenmenger syndrome) are the most common subgroups; others include portopulmonary, HIV, and drug/toxin-induced PAH.
Group 2	PH associated with increased pulmonary capillary wedge pressure (post-capillary PH), due to left-sided heart disease.
Group 3	Lung disease and chronic hypoxia-related PH.
Group 4	Predominantly chronic thromboembolic PH (CTEPH).
Group 5	PH caused by multisystemic disorders or multiple/unknown mechanisms, including sarcoidosis and haematological conditions.

Definition	Characteristics at RHC		
	mPAP (mm Hg)	PCWP (mm Hg)	PVR (WU)
Pre-capillary pulmonary hypertension	≥20	≤15	≥3
Isolated post-capillary pulmonary hypertension	≥20	>15	<3
Combined pre- and post-capillary pulmonary hypertension	≥20	>15	≥3

Table 2 Variables used to predict mortality in pulmonary arterial hypertension and intensify treatment accordingly. This can be adapted to perioperative risk assessment to target preoperative PAH therapeutic optimisation to the 'low risk' column. RA, right atrial; RV, right ventricular; LV, left ventricular; F RV, ejection fraction; RVESVi, RV end-systolic volume index; LVEDVi, left ventricular end diastolic volume index; $RVESvO_2$, mixed venous oxygen saturations; CPET, cardiopulmonary exercise test; CI, cardiac index; RAP, right atrial pressure; 6MWD, 6 min walk distance; VO_2, oxygen uptake; BNP, brain natriuretic peptide; NT-proBNP, N-terminal BNP.

	Risk		
	Low	**Intermediate**	**High**
Clinical assessment			
Right heart failure	None	None	Present
Progression of symptoms	None	Slow	Rapid
Syncope	None	Occasional	Recurrent
Chest pain	None	Occasional	Recurrent
Arrhythmia	None	Occasional	Recurrent
WHO functional class	I/II	III	IV
Imaging and haemodynamics			
Echocardiographic	Preserved RV function. RA area <18 cm^2 No pericardial effusion	Impaired RV function. RA area 18–26 cm^2 No or minimal pericardial effusion	Impaired RV function. RA area >26 cm^2 Pericardial effusion present
Cardiac magnetic resonance	High RVEF (>54%) Normal RVESVi Normal LVEDVi	Reduced RVEF (37–54%) Increased RVESVi Decreased LVEDVi	Reduced RVEF (<37%) Increased RVESVi Decreased LVEDVi
Right heart catheterisation	RAP <8 mm Hg CI >2.5 L min^{-1} m^{-2} SvO_2 >65%	RAP 8–14 mm Hg CI 2.0–2.4 L min^{-1} m^{-2} SvO_2 60–65%	RAP >14 mm Hg CI <2.0 L min^{-1} m^{-2} SvO_2 <60%
Exercise capacity			
6MWD	> 440 m	165–440 m	< 165 m
CPET	Peak VO_2 >15 ml min^{-1} kg^{-1} (>65% predicted) VE/VCO_2 slope <36.0	Peak VO_2 11–15 ml min^{-1} kg^{-1} (35–65% predicted) VE/VCO_2 slope 36.0–44.9	Peak VO_2 <11 ml min^{-1} kg^{-1} (<35% predicted) VE/VCO_2 slope >45.0
Biomarkers			
BNP	< 50 ng L^{-1}	50–300 ng L^{-1}	>300 ng L^{-1}
NT-pro BNP	<300 ng L^{-1}	300–1400 ng L^{-1}	>1400 ng L^{-1}

bilateral lung) remains the ultimate treatment option in PAH when other treatments fail.

In the UK, patients with suspected PH are assessed in nationally commissioned centres to establish the diagnosis and appropriate management. Patients with PAH and CTEPH are rare, with the prevalence of PAH <50 cases per million. However, as patients are living longer with PH, thanks to expert care and advanced therapies, co-morbidities are becoming more prevalent and patients are increasingly presenting with surgical or medical emergencies.[10]

Pathophysiology of perioperative RV failure

Adequate coupling of the RV to the pulmonary circulation is essential for maintaining cardiac output in patients with PH. Most patients with severe PH exhibit significant RV maladaptation or 'uncoupling' to the pulmonary circulation, with a dilated impaired RV, and high RV filling pressure.[11] Right ventricular to pulmonary artery coupling and the response of the cardiopulmonary unit to peri-procedural stressors is an important determinant of perioperative outcome in PH.[12]

Progressive PH leads to a reduction in coronary blood flow, coronary blood flow reserve, and reduced myocardial capillary density, and potentially reduced available oxygen tension in the blood.[13,14] Repetitive cycles of RV ischaemia promote ventricular maladaptation and uncoupling.[15]

Homeometric autoregulation describes maintenance of RV systolic function in response to elevated pulmonary vascular resistance (PVR), an intrinsic adaptation that enables the RV to maintain stroke volume without having to depend on the Frank–Starling mechanism.[16] However, intraoperative interventions that either compromise RV homeometric autoregulation (e.g. sympathetic block such as spinal anaesthesia, thoracic epidural, vasovagal stimuli, or anaesthetic drugs with a negative inotropic effect) or that disproportionately decrease the compliance in small pulmonary vessels (e.g. microembolism, positive pressure ventilation, pulmonary artery clamping, or cytokine storm after tourniquet release) can precipitate RV failure and sudden death.[17–21]

Load-dependent changes in RV function can also impact negatively on cardiac output. An excessive increase in RV preload (e.g. fluid administration, fluid redistribution, or blood transfusion) can increase wall stress and endocardial ischaemia, and lead to severe tricuspid regurgitation and systolic RV dysfunction.[22,23] A disproportionate decrease in preload to a hypertrophied and fibrotic RV with diastolic dysfunction (e.g. vagal overactivity, bleeding, over-diuresis, vasodilatation, impaired venous return owing to increased intra-abdominal pressure, and vena cava compression) can cause decompensation and a low cardiac output state.[24,25] Cardiac output should therefore be optimised perioperatively by controlling RV preload and afterload, supporting RV function to ensure adequate myocardial perfusion with inotropic support, and avoiding or anticipating perioperative insults (Table 3).

Perioperative outcomes in patients with pulmonary hypertension

In a US coding study of more than 140 000 patients with PH from 18 million non-cardiac surgeries between 2004 and 2014, major cardiac events and death occurred in 4–8% of those with PH, compared with 1–2% in those without PH ($P<0.001$).[1] In patients coded as *PAH*, perioperative mortality was 8.3%, myocardial infarction occurred in 4.3%, and stroke in 1.9%.[26] In 12 studies that included patients with PAH and CTEPH (Table 4) undergoing 'major' or 'minor' procedures (according to body cavity insult), 67% of patients had invasive haemodynamic assessment. Perioperative mortality in elective surgery was 3–18%,[3,27] with deaths mostly relating to RV failure occurring within 48 h of surgery. In the only prospective multicentre study, in which 75% of procedures were performed in specialist PH centres, overall mortality was 3.5%. A large US PH centre reported 196 non-cardiac surgery cases (2010–2017) of which 45% were intermediate- to high-risk procedures, mostly performed under general anaesthesia using arterial line pressure monitoring in 55%, pulmonary artery catheter monitoring in 6%, pulmonary vasodilators in 84% and vasopressors in 82% of patients. Major complications such as RV failure and severe arrhythmias were seen in 27%. Overall, 30 day mortality was 3% in elective surgery and up to 15% for emergency surgery. Compared with a previous study by the same group in 2005, mortality fell from 7% to 3%.[4]

No studies report on the specific perioperative risks in patients with Group 2 or 3 PH. However, patients in these PH subgroups were included in cohort and coding studies.[3–5,7] Available data suggest that their overall perioperative risk is increased, but less than in patients with PAH.[1]

Procedure-specific and patient-related risks

Perioperative risk in patients with PH is influenced by patient factors, including severity of PH and RV dysfunction, exercise performance, and functional status; and surgical factors including the urgency, length, and type of surgery (Table 4). From these reports, the estimated mortality in patients with PAH/CTEPH undergoing general surgery is 2–18%, liver transplantation surgery is 35%, and up to 7% for orthopaedic surgery.[4,32,33] Mortality after emergency surgery is significantly higher (15–50%).[3,7,27] US coding studies compared complications, including death, in patients with PH with those without PH in various types of surgery: neurosurgery, 13.0% *vs* 4.1% ($P<0.001$); thoracic surgery, 12.2% *vs* 5.6% ($P<0.001$); vascular surgery, 11.2% *vs* 7.3% ($P<0.001$); and general surgery, 9.3% *vs* 2.7% ($P<0.001$).[26]

Orthopaedic surgery

Lower limb joint surgery poses a risk of intraprocedural pulmonary embolisation of bone marrow, cement, and bone debris causing an increase in PVR and acute RV failure.[34,35] Coding studies report a four-fold increased adjusted risk of mortality (2.4% *vs* 0.6%) for patients with any type of PH undergoing total hip replacement; this was even higher in the PAH group (5% mortality; 95% confidence interval [CI], 2.3–7.7%).[32] Ramakrishna and colleagues[4] followed 145 patients with well-characterised PH and reported a 7% 30 day mortality in patients undergoing orthopaedic surgery. Similarly, Seyfarth and colleagues[36] reported 16 patients on oral PAH therapies undergoing major or joint replacement surgery, in which four underwent general anaesthesia alone, eight combined general anaesthesia with regional, and four regional alone, with one death (6.3%) and six (38%) major perioperative complications. The Sheffield PH centre in the UK recently reported on 14 patients with PAH/CTEPH undergoing joint replacement procedures (10 under regional anaesthesia and

Table 3 Potential perioperative effects on the right ventricle and the pulmonary circulation. PVR, pulmonary vascular resistance; CO_2, carbon dioxide; GA general anaesthesia.

Perioperative event	Heart rate	RV contractility	Pulmonary vascular resistance
Premedication/sedation, peripheral nerve block for minor procedures, or both	Heart rate is often influenced by a high sympathetic drive that can be compromised when cortical activation is reduced with sedatives (i.e. benzodiazepines)	Neutral/mildly reduced	Neutral/mildly increase (+) if CO_2 retention or hypoxaemia present. Patients with sleep apnoea are at particular risk Consider use of high flow oxygen (including Optiflow)
Anaesthetic induction	Decreased. Almost all anaesthetic agents (propofol, thiopental, benzodiazepines) reduce the sympathetic drive to some degree. Ketamine may provide more stability but can precipitate tachycardia or arrhythmias	Neutral or decreased. Highly influenced by PVR and coronary flow	Increased (++) once spontaneous ventilation is inhibited. Hypoxaemia, sympathetic stimulation (laryngoscopy), hypercapnia, high ventilatory pressures all increases the risk
Spinal or epidural anaesthesia and other neuraxial blocks	Sympathetic block	Neutral or decreased because of systemic hypotension; inhibited homeometric regulation	Neutral or reduced, unless stress triggers pulmonary vasoconstriction (+) Use vasopressors to counteract predictable decrease in systemic vascular resistance
Airway management including during extubation	Increased risk of tachyarrhythmia and decompensation	Neutral/increased	Increased
Knife to skin	Increased or decreased independently of pain control – autonomous reflex may trigger vasovagal bradycardia/asystole	Neutral or Increased. Diathermy can affect pacing and biventricular synchrony	Neutral or increased. Poor pain control as a major trigger to increase PVR
Lower limb cementing and tourniquet release	Neutral or initial bradycardia. Tachycardia as a compensating mechanism if RV dysfunction	Neutral. Increased (+) if homeometric regulation is intact. Decreased (++) as a result of limb ischaemic metabolites or failure to overcome increasing afterload	Increased (+++) More relevant during pressurised cementing
Pneumoperitoneum	Neutral – risk of bradycardia/asystole	Reduced	Increased ++
Caesarean section	Increases (++) in awake spinal. Bradycardia if spinal reaches mid-thoracic level or after GA	Decreased (–) volume overload because of fluid shifts (notable 500 ml autotransfusion after delivery)	Decreased (–) once uterus is decompressed. Increases (++) with microembolism during placenta removal.

Table 4 Studies of perioperative morbidity and mortality in patients with pulmonary hypertension undergoing general non-cardiac non-obstetric surgery. ASA, ASA physical status; WHO, World Health Organisation functional class; PoPH, portopulmonary hypertension; RHC right heart catheterisation; PVR, pulmonary vascular resistance; mPAP, mean pulmonary artery pressure; CTEPH, chronic thromboembolic PH; 6MWD, 6 min walk distance; GA/RA, general/regional anaesthesia; RV, right ventricular; PASP, pulmonary arterial systolic pressure; RVSP, right ventricular systolic pressure; Echo transthoracic echocardiography; VATS, video-assisted thoracoscopic surgery; N_2O, nitrous oxide; THR, total hip replacement; TKR, total knee replacement; NYHA, New York Heart Association; PCWP, pulmonary capillary wedge pressure; OR, odds ratio; 95% CI, 95% confidence interval; iPAH, idiopathic PAH; WU, Wood unit; AE, adverse effects; PE, pulmonary embolism; SSc, scleroderma.

Authors, journal, year, (study period) PH centre, Y/N	Patients; type of PH/ PAH	PH severity (RHC/ echo; WHO; 6MWD)	Procedure details	Anaesthesia type	Major complications:	AEs related to preoperative PH severity	Other associations
Mixed general surgical procedures							
Ramakrishna, *JACC*, 2005, (1991–2003) PH centre, Y	145 PH (including 45% PAH (20% iPAH)); 19% lung disease; 8% CTPEH Group 2 PH excluded	RHC RVSP 68 (22) mm Hg (n=68); mPAP 44 (15) mm Hg (n=66); PCWP 15 (8) mm Hg (n=65) Echo RVSP 68 (21) mm Hg (n=145) 73% WHO I–II 27% III/IV	Low & intermediate to high risk	All GA (RA excluded)	Mortality 10% (60% with respiratory failure; 50% with RV failure) Morbidity 42%	Yes: Baseline echo and ECG indices of worse RV function (for mortality but not morbidity) Morbidity related to NYHA <II; 6MWD <330 m	Morbidity predictors (multivariate): history of PE; NYHA >II; intermediate-high risk surgery; duration of anaesthesia >3 h Other mortality predictors (univariate): anaesthesia without *nitrous* oxide (N_2O); use of epinephrine and dopamine
Minai, *Conn Med*, 2006 (1996–2002)[28] PH centre, Y	28 procedures in 21 patients (11 iPAH, 3 SSc, 2 PoPH 5 others)	mPAP 53 (14) mm Hg 38% WHO II; 62% WHO III–IV	Included 12 abdominal; 6 thoracic (including 3 VATS); 4 gynaecologic 4 orthopaedic	GA (22); spinal (4); epidural (1); local (1)	18% mortality Morbidity 14%	No data	No
Kaw, *Respir Med* 2010 (2002–2006) PH centre, Y	Compared 96 with PH to 77 without PH undergoing non-cardiac surgery	12.5% PAH, 40% Group 2 PH, 48%/ mixed PH′	All non-cardiac procedures excluding those under sedation or LA	GA or RA	27 morbid events; 1 patient died (had severe PAH with mPAP >45 mm Hg)	PH predicted complications including heart failure, sepsis, haemodynamic instability, renal failure	Shorter ICU stay, hospital length of stay, Less readmissions in patients without PH (including PAH and Group 2, others)
Lai, *Br J Aanaesth*, 2007 (1999–2004) (54)	62 PH: 18% PAH; 44% Group 2 PH; 21% Group 3 PH	Echo: Mean PASP 79 (9.4) mm Hg (no RHC data)	40 intermediate; 22 low risk procedures (none 'high risk)	GA (36), neuraxial blocks (21)	Mortality 9.7% Morbidity 24%	Yes: significant associations with preop PASP (no data on RV function or 6MWD)	Emergency surgery, higher-risk surgery and coronary heart disease (all P<0.05)
Price, *Eur Respir J*, 2010 (2000–2007) (55) PH centre, Y	28 PAH + non-operable CTEPH	RHC: mean mPAP 43 (12) mm Hg; PCWP 8.8 (3) mm Hg, CI	57% 'major' & 43% 'minor': 16 abdominal	GA (14), RA (14)	Mortality 7% (RV failure) Morbidity 29%	Yes: suggested for exercise related data (shorter	Perioperative complications associated with

Continued

Table 4 Continued

Authors, journal, year, (study period) PH centre, Y/N	Patients; type of PH/ PAH	PH severity (RHC/ echo; WHO; 6MWD)	Procedure details	Anaesthesia type	Major complications:	AEs related to preoperative PH severity	Other associations
		3.25 (0.68) L min^{-1} m^{-2}, PVRI 11.1 (5.7) WU m^{-2}, 75% WHO I–II, 25% WHO III	surgery; 8 orthopaedic; gynaecologic, mastectomy.			6MWD and NYHA III–IV did worse (trends: P=0.058; P=0.14) but not for resting haemodynamic data	emergency surgery (P<0.001); major surgery (P<0.05), long operative time (P<0.05). Use of GA > RA (P=0.12)
Meyer, *Eur Respir J*, 2013 (2007–2010) 75% in PH centre	114 (30% male) IPAH (57, 50%) HPAH (2, 1.8%) Associated PAH (55, 48%) 43% in WHO FC III/IV Most on PAH therapies	RHC RAP 7 (3–10) mm Hg, mPAP 45 (38–55) mm Hg, CO 5 (4.2–5.9) L min^{-1}, CI 2.9 (2.5–3.5) L min^{-1} m^{-2}, PVR 7.2 (4.6–10.5) WU, SvO_2 66% (63–70%)	Abdominal (39%), gynaecological/ urology (22%), trauma/orthopaedic (18%)	GA (82%), Spinal (18%)	Major complications in 6.1%. Mortality 3.5% overall, 2/13 (15%) in emergency procedures	Yes. High RAP (OR=1.1, 95% CI 1–3, P=0.01), 6MWD <399 m (OR=2.2; 95% CI 1.1–3.7, P=0.04), use of perioperative vasopressors, emergency surgery (OR=2.4, 95% CI, 1.4–3.6, P=0.01)	The first prospective multicentre study
Deljou, *J Cardiothorac Vasc Anesth*, 2019 (2010–2017) PH centre, Y	196 PH patients with Group 1 (144, 74%), Group 3 (27, 14%) and Group 4 (25, 13%) 51% on a PDE5i 38% on ERA 1% on PGI2	WHO FC I/II 59% WHO FC III/IV 41% 6MW>330 m in 59% 6MW<330 m in 29% RHC data: RAP >10 mm Hg in 23% mPAP >55 mm Hg in 20%	108 (55%) low-risk, 88 (45%) intermediate- or high-risk procedures.	98.5% GA Pulmonary vasodilators used in 84% cases Vasopressor infusions used in 82%	13 (6.6%) emergencies where 8 (62%) major complications, 2 (15%) deaths Non-emergency group (183 procedures): 45 (25%) complications, 3 (2%) deaths	WHO III/IV *vs* WHO I/II predicted complications (OR 2.72, 95% CI 1.28–5.8), P=0.009 Type of surgery and NTproBNP >300 ng ml^{-1} predicted complications	The most recent study in well characterised PAH in a PH centre
Bennett, *J Clin Anesth*, 2014[31]	33 Eisenmenger patients	PAH	53 procedures	53 GA, RA and/or monitored anaesthetic procedures	Significant hypotension in 14 (26%), oxygen desaturation in 9 (17%) 30 day mortality 3.8% (2 patients)	IV vasopressor use reduced hypotension at induction. Propofol caused most hypotension at induction. Inhalation induction, more likely to result in hypotension (60% of pts)	Both patients that died had minor elective procedures
Orthopaedic surgery							
Memtsoudis, *Anesth Analg*, 2010 (1998–2006) PH centres – not all	US coding: 2184 (0.3%) and 1359 (0.4%) patients with PH out	No RHC details. 17.8% 'primary PH' (call it PAH) of	THA and TKA surgery. In-hospital mortality primary outcome.	Not recorded	In PAH, mortality was 5.2% after THA, 2.3% after TKA	Presence of PH (coding) increased risk of death from THR 4x, and TKR	The first large US coding orthopaedic only study. Only

Continued

Table 4 Continued

Authors, journal, year, (study period) PH centre, Y/N	Patients; type of PH/ PAH	PH severity (RHC/ echo; WHO; 6MWD)	Procedure details	Anaesthesia type	Major complications:	AEs related to preoperative PH severity	Other associations
	of 670.516 undergoing TKA and THA	TKA 19.9% of TKA				4.5x. Patients with idiopathic PAH undergoing THA experienced the highest mortality rate (5% [95% CI, 2.3–7.7%]).	capturing short perioperative period
Seyfarth, *GMS Interdiscip Plast Reconstr Surg DGPW*, 2015 (2011–2014)	16, 6 males iPAH (8), CTEPH (3), Group 3 PH (5)	RHC mPAP 39.3 (6.1) mm Hg WHO II (3) WHO III (11) WHO III–IV (1) All on PAH therapies (9 monotherapy, 3 dual therapy) 2 iloprost, one triple therapy	Major orthopaedic surgery. THR (5), TKR (5), vertebroplasty, spinal fusion (3)	GA alone (6), GA with regional block (6), spinal alone (4)	Morbidity 37.5% (6 patients) with bleeding (n=1), infection (n=1), wound healing defects (3), dysrhythmia (1)	Death (1, 6.25%) caused by lung infection, bradycardia, right heart failure The death was in the iPAH patient with mPAP 47 mm Hg on triple therapy in WHO FC 3–4 preoperatively	The first single-centre orthopaedic only study
Liver transplantation surgery							
Krowka, *Liver Transpl*, 2000 (1996–2000)[29] PH centre, not all	43 patients with PoPH undergoing orthotopic liver transplantation (OLT) (PAH was untreated)	Pre-OLT RHC: mPAP ≥35 mm Hg in 29/ 43 (67%) patients	Liver transplant surgery.	OLT	Morbidity not recorded	15 patients died, 14/ 43 related to cardiopulmonary causes post OLT. 2 deaths intraop. All 6 with severe PAH (mPAP >50 mm Hg) died. 8/23 (35%) with mPAP 35–50 mm Hg died. No deaths if mPAP <35 mm Hg	PVR 5.5 (2.2) *vs* 3.3 (1.9) WU (P<0.005) also predicted cardiopulmonary mortality
Diaz-Gomez, *South Med J*, 2018 (2006–2016)[30] PH centre, Y	75, 78% female Mean age 62 y 86% WHO III–IV	Most were PAH RHC data 97% had ASA risk 3 and 4	37 non-cardiac surgical procedures	Most complications related to GA (70%)	Morbidity 27% 70% of complications related to GA 2 (5%) deaths		Multidisciplinary approach in place.

four under general anaesthesia); 60% received perioperative prostacyclin.[37] Patients were managed postoperatively in a high dependency unit with PH specialist input, and no deaths were reported.

Laparoscopic surgery

Laparoscopic surgery requires carbon dioxide (CO_2) insufflation of the abdominal cavity through a trocar to an intra-abdominal pressure of 12–15 mm Hg. Increasing intra-abdominal pressure can reduce venous return and cardiac output. This may be compounded by the effects of upward diaphragmatic displacement and head-down Trendelenburg positioning on lung compliance and functional residual capacity. Rapid insufflation of CO_2 can also cause a (usually benign) vagal bradycardia, which, in patients with PH, can lead to RV distension, impeded right coronary flow, and cardiac arrest.[18,38,39] Finally, within 5 min of CO_2 insufflation, the increase in end-tidal CO_2 may require an increase in minute ventilation or abdominal deflation to avoid an increase in PVR. Patients with PH are therefore at significant risk of RV/PA uncoupling during laparoscopic surgery through several mechanisms. Potential solutions include minimal and slow CO_2 insufflation, and early conversion to open surgical techniques. There are no studies comparing laparoscopic *vs* open procedures in patients with PH, but there are clear benefits of laparoscopic over open surgery (e.g. pain, bleeding) that should be considered when planning any procedure.[40]

Liver transplantation

Up to 5% of patients on the liver transplant waiting list develop portopulmonary hypertension (PoPH; characterised by an increase in mPAP and PVR), which leads to RV failure and death if not treated with pulmonary vasodilators or liver transplantation. Given that the mortality at 1 yr after liver transplantation is two-fold in patients with PoPH (15–35%) compared with those without PoPH (7–15%),[41] patients with a mPAP >35 mm Hg are now deemed unsuitable for orthotopic liver transplantation unless they respond to PAH therapies.[41,42] However, there is ongoing debate on whether PoPH should be considered an indication rather than contraindication for liver transplant, and whether PVR and RV function should be considered together with mPAP when assessing patients, to account for the increase in cardiac output often seen in liver disease, which can cause an increase in mPAP with a normal PVR.[43]

Perioperative management pathway

No widely accepted protocols exist detailing the perioperative management of patients with PH. Our aim is to provide a protocol-driven structured pathway and to suggest questions for future research. Based on consensus (Fig. 1), a core team (LP, KD, GM, SJW, AB) generated a list of 11 relevant clinical questions and actions and obtained feedback and consensus from co-authors (questions and actions are listed in Fig. 2). Level of evidence was assessed for each recommendation made. The recent European Society of Cardiology guidelines for PH[8] graded evidence as grade A (data derived from multiple randomised clinical trials or meta-analyses), B (data from a single randomised trial), or C (consensus of expert opinion and/or small studies, retrospective studies, registries). The recommendations in this field are all based on 'level' evidence.

Preoperative planning

Perioperative planning and the indications and risks related to surgery should be discussed in a multidisciplinary environment, including senior members of the PH team, anaesthetic team, the relevant operator, and intensivists, including the extracorporeal membrane oxygenation (ECMO) team as indicated (Fig. 2). This ensures that all aspects of care are optimised and risks are minimised.

Recommendation 1. Given the significant risk of complications in patients with PH undergoing general anaesthesia for surgery, also for minor procedures under conscious sedation, cases should be discussed in an multidisciplinary team meeting in a timely fashion. Non-surgical or local/regional anaesthesia alternatives should be pursued if possible.

Preoperative risk assessment

After elective non-cardiac surgery, patients with PH face high rates of fatal complications (3–18%), and even higher for urgent surgery, as summarised above and in Table 4. This far exceeds the background adverse cardiac event rate after non-cardiac surgery of 3.9%.[44] In addition to PH-related risk factors, patients with PH often have comorbidities that contribute to overall risk assessment.

Patient-related risk factors for death include high ASA physical status, WHO functional class III and IV,[3,4,27,45] 6 min walk distance (6MWD) <400 m, and indicators of poorly controlled PH including high brain natriuretic peptide (BNP), RV dysfunction, RAP>7 mm Hg, and high preoperative mPAP, PVR, or both.[3,7] These are similar to thresholds identified in large haemodynamic PAH risk assessment studies of RV function and RV-PA coupling[46]; cardiac magnetic resonance (CMR) indices are increasingly useful.[47] Cardiopulmonary exercise testing (CPET) has an increasing role in the diagnosis and prognosis of patients with PAH.[48] CPET variables, particularly peak oxygen uptake (VO_2peak), end-tidal carbon dioxide tension ($PETCO_2$), and the ventilatory equivalents for CO_2 (minute ventilation/carbon dioxide production relationship [VE/VCO_2]) have emerged as indicators of both PAH severity and prognosis.[48] Furthermore, CPET is now routinely used preoperatively to risk stratify patients before major non-cardiac surgery.[49] Case cohort studies of major non-cardiac surgery including thoracic resections where Group 3 PH may be prevalent, report that reduced VO_2peak and abnormal VE/VCO_2 are associated with increased postoperative morbidity and mortality.[49] CPET may consequently have a role in the preparation of patients before surgery, both in the evaluation of severity of PAH and potentially as a method of perioperative risk stratification. Prospective studies are needed to evaluate the utility of CPET to objectively evaluate functional capacity and gas exchange abnormalities before elective surgery. Procedure-related risk factors include emergency surgery (15–50% mortality), 'major' surgery,[5,7] and long (>2–3 h) operative time (Table 4).[5]

The impact of comorbidities on outcome for patients with PH is not accounted for in available perioperative outcome studies,[4,5,7,27] but is likely to be important. Increasing age is known to increase postoperative cardiac events.[50] Previous studies in patients with idiopathic PAH (iPAH) show that comorbidities including lung disease and smoking history worsen survival.[51] In addition, patients with PAH have associated diseases that may adversely affect surgical healing (e.g. some patients with scleroderma or patients receiving corticosteroids[52]) or conditions increasing the risk of postoperative

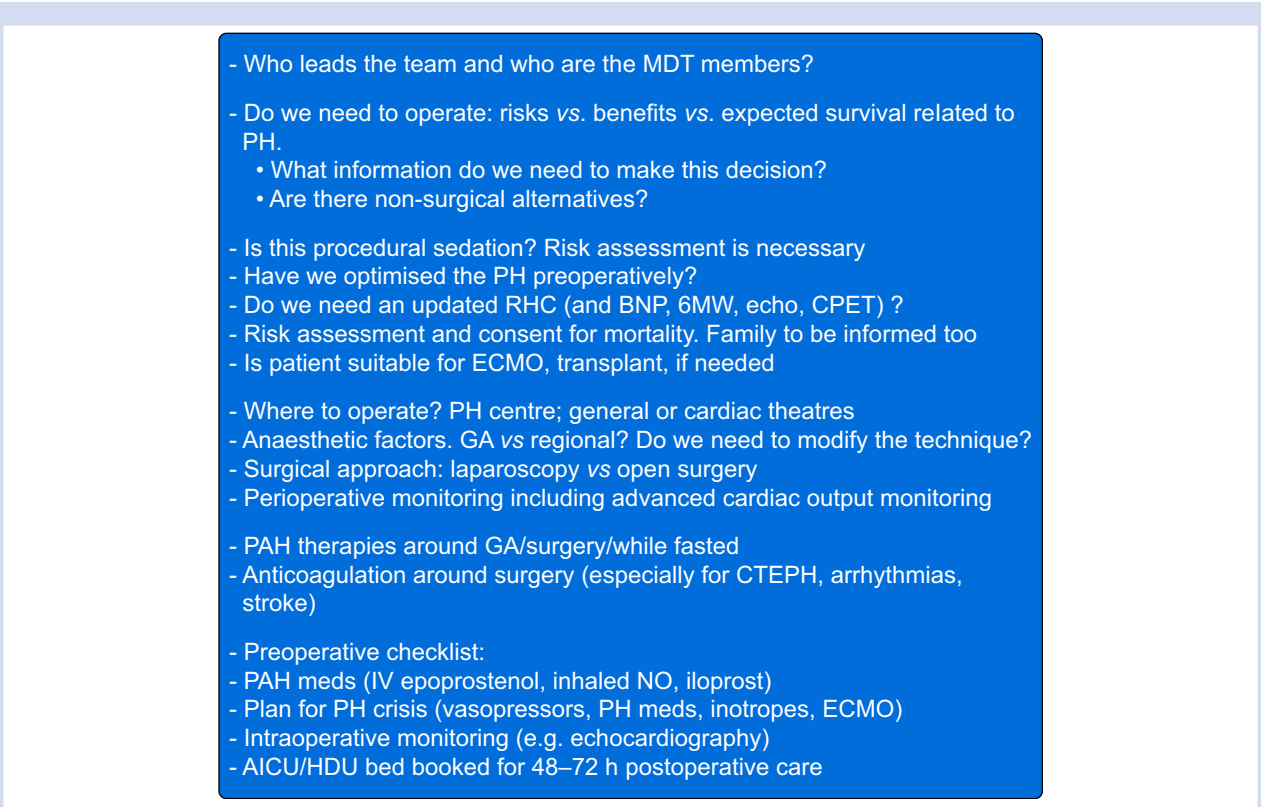


Fig 2. Preoperative questions to consider for patients with pulmonary hypertension undergoing surgery. The following clinical questions are considered in the text: (1) preoperative planning; (2) preoperative risk assessment; (3) where to perform surgery; (4) informed consent; (5) perioperative optimisation of PAH therapies; (6) perioperative monitoring; (7) intraoperative management; (8) postoperative management; (9) management of PH crises; (10) extracorporeal membrane oxygenation (ECMO), transplantation, and cardiopulmonary resuscitation (CPR); (11) how to manage sedation for minor procedures in patients with PH. MDT, multidisciplinary team; PH, pulmonary hypertension; PAH, pulmonary arterial hypertension; CTEPH, chronic thromboembolic PH; RHC, right heart catheterisation; 6MW, 6 min walking test; CPET, cardiopulmonary exercise test; GA, general anaesthesia; NO, nitric oxide; TOE, transoeseophageal echocardiography; AICU, adult ICU; HDU, high dependency unit; BNP, brain natriuretic peptide.

pneumonia (e.g. COPD, lung fibrosis, or obstructive sleep apnoea).[53,54]

Recommendation 2. To formulate an accurate preoperative risk assessment, clinical assessment should include a detailed assessment of RV function and RHC in selected patients. An important role for CPET in this setting is likely to emerge. Every attempt should be made to minimise procedure-related risk factors (Table 3), and optimise pulmonary haemodynamics/RV function according to the 'traffic light' European Society of Cardiology/European Respiratory Society (ESC/ERS) guidance (Table 2), and maximising treatment of comorbidities before surgery.

Where should surgery on patients with PH be performed?

Most reported cases of surgery in well-characterised patients with PAH/CTEPH were performed in specialist PH centres (Table 4). In the only prospective multicentre study, 75% of procedures were performed in specialist PH centres: outcomes were better than outside such centres.[7,55]

Recommendation 3. Planned procedures should be performed in centres with PH specialist support, perioperative anaesthetic, and postoperative intensive care PH expertise. When timely transfer to a PH centre for emergency surgery cannot be achieved, local teams should seek urgent support from a specialist PH centre.

Predicting risk and informed consent

Although the presence of PH is recognised in US guidelines to be a risk factor for postoperative complications,[56] no scoring systems exist that specifically predict perioperative risk in patients with PH. Individualised risk assessment is therefore important.

Recommendation 4. Patients with PH who require a surgical intervention or procedure should be counselled about the risk of perioperative complications including death, need for advanced cardiac support including ECMO (see below), and ceilings of care. Outside an emergency setting, patients should be given sufficient time to formulate their decision, and an additional follow up consultation offered.

Perioperative optimisation of pulmonary vasodilator therapies

Escalation of pulmonary vasodilator therapies may be required in selected cases. PH therapies should be

optimised rapidly in incident cases. In some cases, escalation to prostacyclins may be needed, even in the short term. A reasonable approach for titrating pulmonary vasodilator therapies to reduce perioperative risk would be the use of targets derived from the ESC/ERS Guidelines (Table 2) and those identified from perioperative studies, which include WHO I/II, 3 6MW >400 m, RAP <7 mm Hg, and mPAP <35 mm Hg.[7,41] Pulmonary vasodilator therapies may need to be administered enterally via nasograstric tube (i.e. crushed tablets) or parenterally. Treatment with pulmonary endarterectomy or balloon pulmonary angioplasty may be indicated in patients with CTEPH before time-sensitive surgery such as cancer resection. Recently, preoperative balloon pulmonary angioplasty was performed in a patient with severe CTEPH before successful knee replacement surgery.[57] Optimisation of diuretics and comorbid conditions is also important, and rehabilitation programmes both before ('prehabilitation') and after the procedure. Perioperative anticoagulation should also be planned carefully.

Recommendation 5. Patients with PH awaiting surgery should be carefully risk-assessed, and pulmonary vasodilator therapies modified, escalated, or both as needed before surgery. Some may need a switch to intravenous alternatives (e.g. sildenafil), or escalation to intravenous or inhaled prostanoids. Therapy should not be interrupted before or after surgery.

Perioperative monitoring

Avoidance of right coronary ischaemia is key. Pre-induction invasive arterial blood pressure monitoring is vital for the continuous assessment of beat-to-beat coronary perfusion pressure and targeting of a pre-determined systemic diastolic blood pressure. Central venous pressure (CVP) monitoring has limited application in predicting left ventricular filling pressure and fluid response in critically ill patients with sepsis or multiorgan failure.[58] However, in patients with PH and preserved LV function, the RV is independently subjected to pressure and volume overload, and CVP monitoring can be useful to assess those changes in RV preload, RV filling pressure and global RV performance, with or without changes in volume status.[23,59] An increase in CVP with systemic hypotension often indicates the onset of RV failure and circulatory collapse.[16] The choice of advanced monitoring such as pulmonary artery catheter, transpulmonary dilution (i.e. LiDCO, PiCCO), or transoesophageal echocardiography (TOE) depends on local availability/expertise. Perioperative TOE can assess beat-to-beat RV function (axial and longitudinal) and ventricular filling, dynamic tricuspid regurgitation, and pulmonary pressures, but requires general anaesthesia and an experienced TOE operator. Perioperative transthoracic echocardiography is also used in some centres, if the type of surgery allows.

Recommendation 6. Invasive haemodynamic monitoring with measurement of CVPs (+/− mixed venous oxygen saturation) and systemic arterial pressures is recommended. Additional advanced monitoring should be considered in selected cases.

Intraoperative management

For anaesthetic induction, moderate to high-dose opioids (e.g. 2–7 $\mu g\,kg^{-1}$ fentanyl) as part of a balanced induction technique are most reliable in terms of limiting tachycardia and decreases in systemic vascular resistance (SVR), although recovery time may be prolonged. Propofol is widely used with a vasopressor to avoid hypotension. Etomidate has haemodynamic stability but is not the induction agent of choice in most centres because of adrenal suppressive effects. Some drugs may increase PVR (e.g. ketamine, nitrous oxide, protamine). However, many studies suggest that the effect of ketamine on PVR is minimal, and it can be counterbalanced by its positive sympathetic effect that increases RV performance and maintains SVR.[60,61]

Use of spinal or epidural anaesthesia may be beneficial in PAH patients to avoid general anaesthesia-related haemodynamic instability, although no comparative studies are available. Neuraxial blockade can, however, cause hypotension, compromising RV myocardial perfusion, and can enhance right-to-left shunting in patients with Eisenmenger syndrome, or PAH patients with a patent foramen ovale (PFO). 'Low-dose' spinal or combined spinal–epidural (CSE) techniques have a lesser effect on SVR and may be safe for elective Caesarean section in patients with PH.[62,63]

Despite careful pharmacological induction, tracheal intubation can lead to haemodynamic instability owing to effects of sedation, hypoxaemia, sympathetic stimulation, changes in intrathoracic pressure, and hypoventilation. A high plateau pressure should be avoided during mechanical ventilation, as this can increase PVR and affect RV function. The use of low tidal volume (5–7 ml kg) and low PEEP is recommended to favour gas exchange and minimise lung injury.[64]

Vasoconstrictors such as norepinephrine, vasopressin, phenylephrine, or metaraminol can effectively maintain coronary perfusion pressure in response to decreasing SVR from induction agents. In a recent study of patients with PH undergoing non-cardiac surgery, a perioperative vasopressor was used in 82% of cases.[3] Norepinephrine (up to 0.5 $\mu g\,kg^{-1}\,min^{-1}$) is a first-line treatment in most centres and improves coronary blood flow in models of RV failure. Low-dose vasopressin may be a particularly useful selective systemic vasoconstrictor in this setting, as an infusion or even a bolus in a PH crisis to augment SVR without adverse effects on PVR.[65–67]

In terms of inotropes, beta1-adrenergic receptor agonists such as dobutamine or dopamine can improve RV performance, before (pre-emptive), after anaesthetic induction, and into recovery; however, tachycardia and arrhythmias can be an undesirable effect. No evidence suggests that one catecholamine is superior to another for PH and RV dysfunction. Inodilators or lusitropes such as milrinone or enoximone are useful, providing diastolic augmentation, but because of their dilatation effect, are often co-administered with a vasopressor.[62]

Recommendation 7. A detailed anaesthetic management plan should be made *a priori* in patients with PH and ideally discussed at a multidisciplinary team meeting. Options for regional anaesthesia should be considered, where possible, to avoid general anaesthesia. Pre-induction invasive arterial haemodynamic monitoring is important, and the later addition of CVP monitoring should be considered for major surgery where significant fluid shifts or bleeding is expected. Vasopressors may be needed to counteract the vasodilating effects of induction agents or neuraxial blockade. In most high-risk cases, an inotrope or an inodilator (with a vasopressor) should also be started before anaesthetic induction and continued postoperatively. Intraoperative ventilation should adopt an 'RV protective'

strategy with low plateau pressures. Adequate analgesia is important and early extubation is desirable.[68]

Postoperative management

Most perioperative morbidity and mortality events relate to RV dysfunction and can occur up to 48–72 h after seemingly uncomplicated surgery. Postoperative/post-procedural invasive monitoring in ICU or high dependency unit (HDU) is therefore important, especially in patients with significant PAH and those undergoing complicated and major surgery. PAH therapies should not be interrupted. Line filters are recommended for patients with right-to-left shunting (Eisenmenger syndrome or other PAH with a PFO). For patients previously fully anticoagulated, this should be recommenced as soon as surgically safe with bridging heparin. Analgesia is important to minimise excessive sympathetic drive with resulting tachycardia and increase in PVR, but without excessive sedation causing hypercapnia and increased PVR. Achieving spontaneous ventilation after surgery is paramount to reduce intrathoracic pressure and RV stress. Thus, early extubation is preferable. Postoperative physiotherapy, prophylaxis for deep venous thrombosis and early mobilisation are also indicated.

Recommendation 8. Routine prolonged monitoring after elective surgery in patients with PH should involve postoperative haemodynamic monitoring for at least 48–72 h on ICU or HDU.

Management of pulmonary hypertension crises

Early recognition of a PH crisis is essential. The onset is characterised by an increase in PVR and RV filling pressures, a decrease in cardiac output, hypotension, and a spiral of cardiovascular collapse. Precipitating factors include induction of anaesthesia (vasodilatation), laryngoscopy, hypoxaemia, positive pressure ventilation, extubation, perioperative inflammation and micro-pulmonary embolisation, arrhythmias, and postoperative fluid shifts (Table 2 [3]). Management steps are outlined below.

1. Treat hypotension aggressively to prevent RV ischaemia, without stopping pulmonary vasodilators. Vasopressors include norepinephrine (up to 0.5 μg kg^{-1} min^{-1}) and low dose vasopressin (up to 0.04 units min^{-1}).[66]
2. Reduce postoperative PVR using inhaled and intravenous pulmonary vasodilators. Inhaled NO is an ideal fast-acting potent agent in a ventilated patient.[69,70] Nebulised agents include prostacyclin (with appropriate sonification), iloprost, and milrinone. Intravenous agents include prostacyclin and sildenafil.[70]
3. Improve RV function by optimising preload (aim CVP, 8–12 mm Hg), contractility, and afterload. Adequate preload may require carefully monitored fluid challenges or diuresis. Suitable inotropes include dobutamine, dopamine, and milrinone (and other type 3 phosphodiesterase inhibitors, inhaled or intravenous).[62] Afterload reduction depends on PVR reduction (point 2).
4. Use of ECMO to support the RV can be used as a bridge to postoperative recovery or transplantation.

Recommendation 9. Patients with PH should ideally be operated on in centres with invasive haemodynamic monitoring, echocardiography, and access to inhaled and intravenous pulmonary vasodilators, vasopressors, and inotropes. PH crises should be anticipated and managed promptly. Selected patients may require ECMO, if available and appropriate (see below).

Cardiopulmonary resuscitation, ECMO, and transplantation in the perioperative setting

Despite advances in therapies, the prognosis of patients with PAH remains poor. Attempts at cardiopulmonary resuscitation in patients with PAH have limited success.[71] Therefore, before planning surgery, the long-term prognosis and appropriateness of advanced resuscitation and other invasive modalities should be considered.

In some patients, veno-arterial ECMO may be an option to support the RV as a bridge to recovery or transplantation after high-risk surgery. Some centres may also have access to percutaneous RV assist device (RVAD) support (e.g. the Protek Duo and Tandem Heart cannulae), which may be instituted with or without an oxygenator depending on individual patient need. Because of the high mortality in patients on extracorporeal life support who do not proceed to transplantation, we recommend that high-risk patients are discussed with their local transplant team before elective surgery.[72,73] In very high-risk cases, the team may consider elective insertion of ECMO vascular sheaths at the time of anaesthetic induction to facilitate rapid transfer onto ECMO.

Recommendation 10. ECMO may be considered in selected patients with PH with perioperative haemodynamic instability. Escalation strategy and ceiling of care should be discussed in advance at the multidisciplinary team meeting.

Sedation for minor procedures, including endoscopy

Even mild sedation can cause hypoxaemia, hypercapnia, and increased PVR in high-risk patients with PH. Bronchoscopic biopsy, transbronchial needle aspiration, and manipulation of the airway, including suctioning, can trigger arrhythmias and cardiovascular complications. Bronchoscopy in most patients with group 3 PH can be performed safely with similar outcomes to non-PH patients[74–76]; however, the onset of hypoxaemia is more common in patients with PH,[77] and the complication rate in patients with more severe PH is unclear.[78] Oxygen use is likely to reduce complication rates during bronchoscopy.[77,79]

There are few studies in patients with severe PH undergoing gastrointestinal and gynaecological endoscopy. Vagal stimulation during these procedures may even precipitate cardiac arrest in patients with severe PH. However, gastrointestinal endoscopy is essential and safer than conservative treatment in patients with PH presenting with gastrointestinal bleeding.[80] Lai and colleagues[81] reported 37 gastrointestinal endoscopic procedures performed in patients with PAH under sedation or general anaesthesia (in 16%), with 24% of cases recovering on ICU after the procedure. Major complications to 28 days occurred in five (14%) cases, including severe hypotension, arrhythmias and RV failure. Although no deaths were reported in this French National PH reference centre study, the complication rate exceeded the comparable cardiopulmonary adverse event rate of 1.8% in patients with ASA physical status 3 reported in a US study of more than 300 000 gastrointestinal procedures.[82]

Recommendation 11. Patients with severe PH requiring endoscopy should be carefully monitored by experienced staff in PH centres.

Anaesthesia and sedation in patients with coronavirus disease 2019

During the recent coronavirus disease 2019 (COVID-19) pandemic, reports describe micro- and macro-pulmonary thrombosis, with RV dysfunction as a common echocardiographic feature.[79] Residual impairment in lung perfusion, alongside resolving parenchymal inflammation, has been observed in recovering patients. The concepts and recommendations discussed in this report may therefore be relevant in patients with or recovering from COVID-19 who require surgery. Careful preoperative assessment and perioperative monitoring with echocardiography is strongly recommended in this patient group.

Conclusions

Risks and benefits should be carefully considered when planning surgical, endoscopic, or other procedures in patients with known PAH or CTEPH. Such invasive procedures should be performed in specialist PH and cardiac centres where a structured perioperative multidisciplinary approach can be taken, with preoperative optimisation, perioperative haemodynamic monitoring, and management of complications, including early recognition and prompt treatment of PH crises. If careful planning and transfer to a specialist PH centre is not possible because of the urgent nature of the procedure, preoperative advice from a PH team should be sought to guide local teams and help avoid pitfalls. Our systematic review highlights the limited available evidence. Important questions for future studies include the role for CPET in risk assessment in this population, the role for exercise training in optimisation, and the risk of borderline PH. National registries would be a useful start to standardise approaches.

Authors' contributions

Study concept and design: LP, GM, KD, SJW
Data acquisition: LP, GM, KD, SJW
Data analysis: LP, GM, KD, SJW, AB, BG, DL, GC, SP, SVK, AR, CS
Data interpretation: TP, DA, TA, MG, CM, PM, MG, CH, AK, DK, RC, LH, RD, BS, JL, DT, SG, MJ, CC, DW, AC, SL
Writing of the initial manuscript draft: LP, GM, KD, SJW, AB, BG, DL, GC, SP, SVK, AR, CS
Critical revision of the manuscript for important intellectual content: TP, DA, TA, MG, CM, PM, MB, CH, AK, DK, RC, LH, RD, BS, JL, DT, SG, MJ, CC, DW, AC, SL.
All authors reviewed the final manuscript, and agree to be accountable for all aspects of the work.

Declarations of interest

LP has received personal fees for educational meetings and/or advisory boards from Janssen and GSK. DGK has received personal fees for participation in educational meetings and/or advisory boards and received funding to attend educational meetings from Janssen, Bayer, GSK and MSD. His department has received research funding from Janssen and GSK. MAG has served on advisory boards of Janssen UK and Global, Pfizer UK, GSK and Bayer, and has received unrestricted educational grants from Janssen Global, GSK and Pfizer UK. DT declares speaker honoraria and educational grants from Janssen, Bayer, GlaxoSmithKline, J&J, Lilly, MDS, Pfizer. SG has had honoraria for advisory boards, speaker fees and / or drug safety boards from Janssen and United Therapeutics. RC has honoraria for advisory boards and speakers fees from MSD and Janssen. LH has received honoraria for speaker fees and / or scientific advice from Janssen, Bayer, GSK and Third Pole and unrestricted education grant from Janssen and GSK. CM has received speaker fees and an educational grant from Janssen within the last three years. BM has received speaker fees from Janssen. AC has received speaker fees and travel reimbursement from Nordic Pharma. BES has received speaker fees and fees for educational meetings and/or advisory boards from Janssen and GSK. JGC has received personal fees for participation in educational meetings and/or advisory boards, Janssen, Bayer, GSK and MSD and received funding to attend educational meetings from Janssen. His department has received research funding from Janssen. KD has acted as consultant and received unrestricted educational grants and speaker fees from Janssen. SJW has received honoraria for advisory boards for Janssen, MSD, honoraria for speaking and travel grants from GSK, MSD, Bayer and Janssen and research grants from Janssen and Bayer.

Appendix A. Supplementary data

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